Module 9: Probability Distributions

Instructional Hours: 4

In Module 8, you were introduced to the concept of random variables and its relevance in your area of study. In this module, you will extend this knowledge and learn more about Discrete and Continuous Probability distributions and their applications.

Discrete and Continuous Probability distributions

- Probability distributions describe how values of a random variable are distributed. Since we can either have discrete or continuous random variables, probability distributions can also be classified as either textitdiscrete or continuous.
- A discrete probability distribution describes a random variable that can take on a finite or countably infinite set of distinct values.
- The probability of each value is represented by a probability mass function (PMF), which is a rule that assigns a probability to each possible outcome.

Properties of Discrete Probability Distributions

- The set of possible values is countable (either finite or countably infinite).
- The probability of each value is given by the PMF, denoted as $P(X = x)$.
- The sum of probabilities for all possible values must equal 1:

$$\sum P\left(X = x_i\right) = 1$$

where x_i are the possible values of the random variable X.

- A continuous probability distribution describes a random variable that can take on any value within a certain interval or range.
- In this case, the probability of a random variable taking any specific value is 0 . Instead, probabilities are represented over intervals of values, and the probability density function (PDF) describes the likelihood of the variable falling within a certain range.

Properties of Continuous Distributions

- The set of possible values is uncountable, typically representing a continuous range.

- The probability of the random variable taking a specific value is 0 , i.e., $P(X = x) = 0$ for all x.
- Probabilities are found by calculating the area under the PDF curve:

$$P(a \leq x \leq b) = \int_a^b f(x)dx$$

where $f(x)$ is the PDF of the continuous random variable X.

Some Common Probability Distributions

Discrete Probability Distributions

Bernoulli Distribution

Bernoulli distribution describes a random experiment that has only exactly two possible outcomes: "success" (often coded as 1) and "failure" (often coded as 0).

Definition. *A Bernoulli trial is a random experiment in which there are only two possible outcomes, which are success or failure.*
For example

- *Tossing a coin and considering heads as success and tails as failure.*
- *Checking items from a production line, where success is not defective and failure is defective.*
- *Calling a call centre, success means operator is free and failure means no operator is free.*

A Bernoulli random variable X takes the values 0 and 1 , and $P(X = 1) = p$ and $P(X = 0) = 1$ - p.

Definition. *A random variable X follows a Bernoulli distribution with parameter p if the probability mass function is given by*

$$P(X = x) = \begin{cases} p^x \cdot (1-p)^{1-x} & \textit{for } x = 0, 1 \\ 0 & \textit{otherwise} \end{cases} \tag{3}$$

This is abbreviated as X Bernoulli(p)

Properties of Bernoulli Distribution

1. Bernoulli distribution is defined for discrete random variables that take on the values 0 or 1.

2. The distribution is characterized by a single parameter p, the probability of success.

3. The probability mass function of a Bernoulli random variable X is given by equation 3.

Expectation and Variance for Bernoulli

.

$$
\begin{aligned}
E(X) &= \sum_{x=0}^{1} x f(x) \\
&= \sum_{x=0}^{1} x p^{x} (1-p)^{1-x} \\
&= 0 \times (1-p) + 1 \times p \\
&= p
\end{aligned}
$$

and

$$
\begin{aligned}
\operatorname{Var}(X) &= E\left(X^2\right) - (E(X))^2 \\
&= p - p^2 \\
&= p(1-p)
\end{aligned}
$$

Example 1. *Suppose you have a fair coin. Let X be a random variable where ($X = 1$) if the coin lands on heads(success) and ($X = 0$) if it lands on tails (failure). Calculate the expectation and variance of (X).*

Solution:

- *Since the coin is fair, the probability of heads (success) if* $p = 0.5$ *and the probability of tails (failure) is* $1 - p = 0.5$

- $E(X) = p = 0.5$

- $\operatorname{Var}(X) = p(1-p) = 0.5 \cdot 0.5 = 0.25$

Example 2. *Consider a biased coin that lands on heads with a probability of 0.7 . Let X be a random variable where $X = 1$ if the coin lands on heads and $X = 0$ if it lands on tails. Calculate the expectation and variance of X.*

Solution:

- *Probability of heads (success) is* $p = 0.7$

- *Probability of tails (failure) is* $1 - p = 1 - 0.7 = 0.3$
- $E(X) = p = 0.7$
- $\text{Var}(X) = p(1-p) = 0.7 \cdot 0.3 = 0.21$

Example 3. *A box contains 5 good oranges and 15 bad ones. Let* $X = 1$ *if the orange is good and* $X = 0$ *if the orange is bad.*
If an orange is selected at random from the box determine
(i) the probability of selecting a good orange
(ii) the mean of X
(iii) the variance of X

Solution:

(i)

$$P(X = 1) = \frac{5}{20} = \frac{1}{4} = p$$

(ii)

$$E(X) = p = \frac{1}{4}$$

(iii)

$$\text{Var}(X) = p(1-p) = \frac{3}{16}$$

Binomial Distribution

Binomial distribution is a discrete probability that describes the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success. This distribution generalizes Bernoulli distribution to multiple trials.

Definition. *A random variable* X *follows a Binomial distribution if it represents the number of successes in* n *independent Bernoulli trials with probability* p *of success in each trial. We*
denote this by

$$X \sim \text{Bin}(n, p)$$

The probability mass function of a binomial random variable X *is given by*

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x \in X = \{0, 1, 2, \cdots, n\} \quad n \geq 1, \quad 0 \leq p \leq 1$$

Properties of Binomial Distribution

1. Binomial distribution is defined for discrete random variables that count the number of successes in a fixed number of trials.
2. The distribution is characterized by two parameters: n - the number of trials and p - the probability of success in each trial.
3. Each trial is independent of the others.
4. The probability of success p remains constant in each trial.

Example 4. *A biased coin is tossed 6 times. The probability of heads on any toss is 0.3 . Let X denote the number of heads that come up. Calculate*

1. $P(X < 2)$
2. $P(X = 3)$
3. $P(1 < X < 5)$

Solution:

1. $P(X = 2) = \binom{6}{2}(0.3)^2(0.7)^4 = 0.324135$
2. $P(X = 3) = \binom{6}{2}(0.3)^3(0.7)^3 = 0.18522$
3.

$$\begin{aligned} P(1 < X \leq 5) &= P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ &= 0.324 + 0.185 + 0.059 + 0.01 \\ &= 0.578 \end{aligned}$$

Example 5. *A quality control engineer is in charge of testing whether or not 90% of the DVD players produced by his company conform to specifications. To do this, the engineer randomly selects a batch of 12 DVD players from each day's production. The day's production is acceptable provided no more than 1 DVD player fails to meet specifi-*
cations. Otherwise, the entire day's production has to be tested.

1. *What is the probability that the engineer incorrectly passes a day's production as acceptable if only 80% of the day's DVD players conform to specification?*
2. *What is the probability that the engineer unnecessarily requires the entire day's production to be tested if in fact 90% of the DVD players conform to specification?*

Solution:

1. *Let X denote the number of DVD players in the sample that fail to meet specifications. We therefore want* P(X $\leq$ 1 *) with binomial parameters* n = 12 *and* p = 0.2*, so*

$$
\begin{aligned}
P(X \leq 1) &= P(X=0) + P(X=1) \\
&= \binom{12}{0}(0.2)^0(0.8)^{12} + \binom{12}{1}(0.2)^1(0.8)^{11} \\
&= 0.069 + 0.206 = 0.275
\end{aligned}
$$

2. *We now want $P(X > 1)$ with parameters $n = 12$ and $p = 0.1$*

$$
\begin{aligned}
P(X \leq 1) &= P(X=0) + P(X=1) \\
&= \binom{12}{0}(0.1)^0(0.9)^{12} + \binom{12}{1}(0.1)^1(0.9)^{11} \\
&= 0.659 \\
P(X > 1) &= 1 - P(X \leq 1) \\
&= 1 - 0.659 = 0.341
\end{aligned}
$$

Expectation and Variance of Binomial Distribution

1. **We start with the expectation.**

- Let $X_1, X_2, \cdots, X_n$ be n independent Bernoulli random variables, each with probability p of success.
- The binomial random variable X can be expressed as the sum of these Bernoulli random variables.

$$X = \sum_{i=1}^{n} x_i$$

- Therefore the expectation of the binomial random variable is

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E\left(X_i\right) = \sum_{i=1}^{n} p = n \cdot p$$

2. **For the variance we have**

- The variance of a Bernoulli random variable X_i with success probability p is

$$\mathrm{Var}(X) = p(1-p)$$

- For independent random variables $X_1, X_2, \cdots, X_n$, the variance of their sum is the sum of their variances

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i)$$

- Since $\operatorname{Var}(X) = p(1-p)$ for each i

$$\operatorname{Var}(X) = \sum_{i=1}^{n} p(1-p) = n \cdot p(1-p)$$

Example 6. *Suppose you flip a coin 10 times. Let X be the random variable representing the number of heads (successes) obtained. Calculate the expectation and variance of X.*

Solution:

- *Number of trials* n = 10
- *Probability of success* p = 0.5*, since the coin if fair.*
- *Expectation is*

$$E(X) = n \cdot p = 10 \cdot 0.5 = 5$$

- *Variance is*

$$\begin{aligned}\operatorname{Var}(X) &= n \cdot p \cdot (1-p) \\ &= 10 \cdot 0.5 \cdot 0.5 \\ &= 2.5\end{aligned}$$

Video Visit the URL below to view a video:

`https://www.youtube.com/embed/eSJ6ufTSJNk`

Binomial Distribution

Poisson Distribution

Poisson distribution is used to model the number of events such as the number of telephone calls at a business, number of customers in waiting lines, number of defects in a given surface area or the number of accidents at an intersection, occurring within a given time interval.

Poisson probabilities are useful when there are a large number of independent trials with a small probability of success on a single trial and the variables occur over a period of time.

Definition. *A random variable X follows a Poisson distribution with parameter* $\boldsymbol{\lambda}$ *- the average number of events in a given interval if*

$$\mathrm{P(X = k)} = \frac{\lambda^{\mathrm{k}}\mathrm{e}^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots \tag{4}$$

where $\boldsymbol{\lambda} > 0$ *is the average rate at which events occur, e is the base of the natural logarithm. This is abbreviated as*

$$X \sim \mathrm{Pois}(\lambda)$$

Remark. *The major difference between Poisson and Binomial distributions is that the Poisson distribution does not have a fixed number of trials. Instead, it uses the fixed interval of time or space in which the number of successes is recorded.*

Properties of Poisson Distribution

1. Poisson distribution is defined for discrete random variables that count the number of events in a fixed interval.
2. It is characterized by a single parameter, $\boldsymbol{\lambda}$, which is the expected number of events in the interval.
3. The probability mass function of a Poisson random variable X is given by equation 4.
4. If $X \sim$ Poisson $(\boldsymbol{\lambda}_1)$ and $Y \sim$ Poisson $(\boldsymbol{\lambda}_2)$ are independent, then $X + Y \sim$ Poisson $(\boldsymbol{\lambda}_1 + \lambda_2)$.

Example 7. *A call center receives an average of 3 calls per minute. What is the probability that exactly 5 calls are received in a given minute?*

Solution:

- *Here,* $\boldsymbol{\lambda} = 3$ *and we need to find* $P(X = 5)$.
- *Using the probability mass function of the Poisson distribution*

$$
\begin{aligned}
\mathrm{P(X = 5)} &= \frac{3^5 \mathrm{e}^{-3}}{5!} \\
&= \frac{243\mathrm{e}^{-3}}{120} \\
&\approx 0.1008
\end{aligned}
$$

Example 8. *Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one minute interval there will be*

1. *0 jobs*
2. *exactly 2 jobs*
3. *at most 3 arrivals*
4. *more than 3 arrivals*

Solution: *The job arrive at an average of* $\boldsymbol{\lambda} = 2$

1. *No job arrivals:* $P(X = 0) = e^{-2} = 0.1353353$
2. *Exactly 3 job arrivals:* $P(X = 3) = \frac{2^3 e^{-2}}{3!} = 0.1804470$
3. *At most 3 arrivals*

$$
\begin{aligned}
P(X \leq 3) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\
&= \left[1 + \frac{2}{1} + \frac{2^2}{2} + \frac{2^3}{3!}\right] e^{-2} \\
&= 0.8571
\end{aligned}
$$

4. *More than 3 arrivals*

$$
\begin{aligned}
P(X > 3) &= 1 - P(X \leq 3) \\
&= 1 - 0.8571 = 0.1429
\end{aligned}
$$

Expectation and Variance for Poisson

Expectation- If X is a random variable following Poisson distribution with parameter $\boldsymbol{\lambda}$, then expectation of X, denoted as $E(X)$ is given by $E(X) = \lambda$

If X is a random variable following Poisson distribution with parameter λ, then variance of X, denoted as Var(X) is given by Var(X) = $\boldsymbol{\lambda}$

Note: For Poisson distribution, E(X) = Var(X) = $\boldsymbol{\lambda}$

Example 9. *In a large book, typographical errors occur at an average rate of 2 per 100 pages. Assume that the number of typographical errors in 100 pages follows a Poisson distribution. Calculate the expectation and variance of the number of typographical errors in 100 pages.*

Solution:

- *In this example, the value of lambda is 2.*
- *So the mean and the variance are equal to 2 .*

Poisson Approximation to the Binomial Distribution

Poisson approximation to Binomial distribution is useful tool when dealing with Binomial distributions where the number of trials is large, and the probability of success in each trial is small. Therefore, to proceed, we assume that n is large and p is small, np = $\boldsymbol{\lambda}$ is finite.

.

- *We start with the binomial pmf*

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$
$$= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

- *Simplify the binomial coefficient for large n. Note that for large n, $\frac{n!}{(n-k)!}$ can be approximated by $\mathrm{n^k}$, as k is relatively small compared to n :*

$$\frac{n!}{k!(n-k)!} \approx \frac{n^k}{k!}$$

- *Since* $\mathrm{p} = \frac{\boldsymbol{\lambda}}{\mathrm{n}}$, *we substitute this into the PMF:*

$$\mathrm{P(X = k)} = \frac{\mathrm{n}^k}{\mathrm{k!}} \left(\frac{\lambda}{\mathrm{n}}\right)^{\mathrm{k}} (1-\mathrm{p})^{\mathrm{n-k}}$$
$$= \frac{\mathrm{n}^k}{\mathrm{k!}} \cdot \frac{\lambda^k}{\mathrm{n}^k} \cdot (1-\mathrm{p})^{\mathrm{n-k}}$$
$$= \frac{\lambda^k}{\mathrm{k!}} (1-\mathrm{p})^{\mathrm{n-k}}$$

- *As* $\mathrm{n} \to \infty$ *and* $\mathrm{p} \to 0$, *the term* $(1-\mathrm{p})^{\mathrm{n-k}}$ *can be approximated using the limit* $\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$, *therefore*

$$(1-p)^{n-k} \approx (1-p)^n = \left(1 - \frac{\lambda}{n}\right)^n \approx e^{-\lambda}$$

- *This is the PMF of the Poisson distribution with parameter* $\boldsymbol{\lambda}$*.*

Video Visit the URL below to view a video:

https://www.youtube.com/embed/RJ7A0xQQCGY

Poisson Distribution

Up to this point, I hope you have enjoyed learning about Basic Concepts in Probability, Random Variables and Discrete Distribution Functions. You are now going to learn about Continuous Probability Distribution Functions.

Continuous Probability Distributions

Normal Distribution

The Normal Distribution, also known as the Gaussian distribution, is a continuous probability distribution defined by the probability density function (PDF):

.

$$\mathrm{f}(\mathrm{x} \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\mathrm{x}-\mu)^2}{2\sigma^2}\right) \tag{5}$$

where:

- μ *is the mean or expectation of the distribution.*
- σ *is the standard deviation, which measures the spread or dispersion of the distribution.*
- x *is the variable for which the probability is being calculated.*

Properties of the Normal Distribution

- Normal distribution is symmetric about the mean $\boldsymbol{\mu}$.
- Mean, median, and mode of the distribution are approximately equal and located at $\mathrm{x} = \boldsymbol{\mu}$.
- The total area under the curve is equal to 1.
- Approximately 68% of the data falls within one standard deviation ($\mu \pm \sigma$), 95% within two standard deviations ($\mu \pm 2\sigma$), and 99.7% within three standard deviations ($\mu \pm 3\sigma$). This is known as the 68-95-99.7 rule.

Standard Normal Distribution

This is a special case of the normal distribution with a mean of 0 and a standard deviation of 1. Its PDF is:

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) \tag{6}$$

where $\mathrm{z} = \frac{\mathrm{x}-\boldsymbol{\mu}}{\boldsymbol{\sigma}}$ is the z -score, which standardizes the variable x by measuring how many standard deviations it is from the mean.

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) of the normal distribution is given by:

$$\mathrm{F}(\mathrm{x} \mid \boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathrm{P}(\mathrm{X} \leq \mathrm{x}) = \int_{-\infty}^{\mathrm{x}} \mathrm{f}(\mathrm{t} \mid \boldsymbol{\mu}, \boldsymbol{\sigma}) \mathrm{dt} \tag{7}$$

We now consider some illustrations of how probability is applied in problems involving normal distribution.

Illustrations on the Use of Normal Distribution

Illustration. *1.* **Probability in a Normal Distribution** *Suppose the heights of adult men are normally distributed with a mean of* $\mu = 70$ *inches and a standard deviation of* $\sigma = 3$ *inches. We want to find the probability that a randomly selected man is shorter than 68 inches.*

The z-score is calculated as:

$$z = \frac{68 - 70}{3} = -\frac{2}{3} \approx -0.67 \tag{8}$$

Using a standard normal table or calculator, we find that:

$$\mathrm{P}(\mathrm{X} < 68) = \Phi(-0.67) \approx 0.2514 \tag{9}$$

Thus, the probability that a man is shorter than 68 inches is approximately 25.14% *or 0.2514 .*

2. **Finding a Height Corresponding to a Given Probability** *In the same distribution of men's heights (* $\boldsymbol{\mu} = 70, \boldsymbol{\sigma} = 3$ *), suppose you want to find the height below which* 90% *of men fall.*
We need to find x *such that* $P(X < x) = 0.90$. *Using the inverse of the standard normal distribution (or a z-table), we know that:*

$$\Phi(z) = 0.90 \Longrightarrow z \approx 1.28 \tag{10}$$

Now we convert the z-score back to the original scale:

$$x = \mu + z\sigma = 70 + 1.28(3) = 70 + 3.84 = 73.84 \tag{11}$$

Thus, 90% *of men are shorter than approximately 73.84 inches.*

Example 10. *Suppose the weights of apples in a farm are normally distributed with a mean of* $\mu = 150$ *grams and a standard deviation of* $\sigma = 20$ *grams. What is the probability that a randomly selected apple weighs less than 130 grams?*

Solution:

Step 1: *Calculate the z -score for* $X = 130$ *using the formula:*

$$\mathrm{z} = \frac{\mathrm{X} - \mu}{\sigma} \tag{12}$$

Substitute the given values:

$$z = \frac{130 - 150}{20} = \frac{-20}{20} = -1 \tag{13}$$

Step 2: *Use the standard normal distribution table or a calculator to find the probability corresponding to* $z = -1$*. The cumulative probability is:*

$$\mathrm{P}(\mathrm{X} < 130) = \Phi(-1) \approx 0.1587 \tag{14}$$

Thus, the probability that a randomly selected apple weighs less than 130 grams is approximately 15.87% *or 0.1587 .*

Example 11. *The heights of adult women are normally distributed with a mean of* $\mu = 65$ *inches and a standard deviation of* $\sigma = 3$ *inches. What is the height below which* 90% *of the women fall?*

Solution

Step 1: *Find the z-score that corresponds to a cumulative probability of 0.90 . From the standard normal table, we find:*

$$\Phi(z) = 0.90 \Longrightarrow z \approx 1.28 \tag{15}$$

Step 2: *Convert the z-score to the original scale using the formula:*

$$X = \mu + z\sigma \tag{16}$$

Substitute the given values:

$$X = 65 + 1.28(3) = 65 + 3.84 = 68.84 \tag{17}$$

Thus, 90% of adult women are shorter than approximately 68.84 inches.

Example 12. *Assume that the IQ scores of a population are normally distributed with a mean of* $\boldsymbol{\mu} = 100$ *and a standard deviation of* $\sigma = 15$. *What is the probability that a randomly selected person has an IQ score between 85 and 115?*

Solution: Step 1: *Calculate the* z *-scores for both* $X = 85$ *and* $X = 115$ *:*

$$z_1 = \frac{85 - 100}{15} = \frac{-15}{15} = -1 \tag{18}$$

$$z_2 = \frac{115 - 100}{15} = \frac{15}{15} = 1 \tag{19}$$

Step 2: *Find the cumulative probabilities for* $z_1 = -1$ *and* $z_2 = 1$. *From the standard normal table:*

$$\Phi(-1) \approx 0.1587 \tag{20}$$

$$\Phi(1) \approx 0.8413 \tag{21}$$

Step 3: *The probability that the IQ score is between 85 and 115 is the difference between these two probabilities:*

$\mathrm{P}(85 < \mathrm{X} < 115) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = 0.6826$

Thus, the probability that a randomly selected person has an IQ score between 85 and 115 is approximately 68.26%.

Example 13. *A machine fills bottles with a mean volume of 500 ml and a standard deviation of 10 ml . If a bottle is selected at random, what is the probability that the bottle contains more than 515 ml of liquid?*

Solution: Step 1: *Calculate the* z *-score for* $\mathrm{X} = 515$ *:*

$$z = \frac{515 - 500}{10} = \frac{15}{10} = 1.5 \tag{23}$$

Step 2: *Find the cumulative probability for* $z = 1.5$. *From the standard normal table:*

$$\Phi(1.5) \approx 0.9332 \tag{24}$$

Step 3: *The probability that a bottle contains more than 515 ml is:*

$$P(X > 515) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668 \tag{25}$$

Thus, the probability that a randomly selected bottle contains more than 515 ml of liquid is approximately 6.68%.

Example 14. *The scores on a standardized test are normally distributed with a mean of 500 and a standard deviation of 100 . What score separates the top* 5% *of test-takers from the rest?*

Solution: Step 1: *Find the z-score corresponding to the top 5% (or 95% cumulative probability). From the standard normal table:*

$$\Phi(z) = 0.95 \implies z \approx 1.645 \tag{26}$$

Step 2: *Convert the z-score to the original scale:*

$$X = \mu + z\sigma \tag{27}$$

Substitute the given values:

$$X = 500 + 1.645(100) = 500 + 164.5 = 664.5 \tag{28}$$

Thus, a score of approximately 664.5 separates the top 5% of testtakers from the rest.

Exponential Distribution

Exponential distribution is used to model the time between independent events that happen at a constant average rate.

Definition. *A random variable X follows an Exponential distribution with parameter* $\boldsymbol{\lambda}$ *- the rate parameter in a given interval if*

$$P(X = k) = f(x; \lambda) = \begin{cases} \lambda \cdot e^{-\lambda \cdot x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{29}$$

where $\boldsymbol{\lambda} > 0$ *is the rate parameter, e is the base of the natural logarithm. This is abbreviated as*

$$X \sim \text{Exp}(\lambda)$$

Exponential distribution is often employed in reliability engineering to model time to failure of systems and components, in queuing theory to model waiting times, and in survival analysis.

Properties of Exponential Distribution

1. The probability mass function of an Exponential random variable X is given by equation 29, and the CDF is given as

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda \cdot x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{30}$$

2. Mean and variance in exponential distribution is given as $E(X) = \frac{1}{\lambda}$ and $\text{Var}(X) = \frac{1}{\lambda^2}$.

3. The distribution has the memoryless property, which means that the probability of an event occurring in the future is independent of any past events.

4. Exponential distribution is often used to model the time between events in a Poisson process, where events occur continuously and independently at a constant average rate. If the number of events in a time interval t follows a Poisson distribution with rate $\boldsymbol{\lambda} \cdot \mathrm{t}$ then the time between consecutive events follows an exponential distribution with rate parameter $\boldsymbol{\lambda}$.

Gamma Distribution

Gamma distribution is an extention of exponential distribution since it allows modeling of the time until the occurrence of α events in a Poisson process.

It is characterized by two parameters: the shape parameter $\boldsymbol{\alpha}$ (which must be positive and typically takes integer or real values) and the rate parameter $\boldsymbol{\beta}$ (also positive).

Gamma distribution is highly versatile and widely used in various fields. Its flexibility makes it suitable for modeling a wide range of phenomena, particularly those involving waiting times. For example, it can describe the time until the next rainfall in meteorology, the total time for a series of business transactions, or the lifetime of electronic components subjected to varying stress conditions. Understanding the Gamma distribution helps in grasping more complex statistical models and applications in real-world scenarios.

Definition. *A random variable X follows an Gamma distribution with parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ if*

$$\mathrm{P(X = k) = f(x;} \boldsymbol{\alpha}, \boldsymbol{\beta}) = \begin{cases} \frac{\boldsymbol{\beta}^{\alpha} \cdot \mathrm{x}^{\alpha - 1} \mathrm{e}^{-\beta x}}{\Gamma(\boldsymbol{\alpha})} & \mathrm{x} \geq 0 \\ 0 & \mathrm{x} < 0 \end{cases} \tag{31}$$

where $\boldsymbol{\alpha} > 0$ is the shape parameter, $\boldsymbol{\beta} > 0$ is the rate parameter. This is abbreviated as

$$X \sim \text{Gamma}(\boldsymbol{\alpha}, \boldsymbol{\beta})$$

Properties of Gamma Distribution

1. The probability density function is given by equation 31 , where $\boldsymbol{\alpha} > 0, \boldsymbol{\beta} > 0$, and $\boldsymbol{\Gamma}(\boldsymbol{\alpha})$ is the Gamma function defined by

$$\Gamma(\alpha) = \int_0^\infty \mathrm{t}^{\alpha - 1} \mathrm{e}^{-\mathrm{t}} \mathrm{dt}$$

2. The mean and the variance are: $\boldsymbol{\mu} = \frac{\alpha}{\boldsymbol{\beta}}$ and $\sigma^2 = \frac{\alpha}{\boldsymbol{\beta}^2}$
3. The Gamma distribution with $\boldsymbol{\alpha} = 1$ is the exponential distribution with rate parameter β.
4. If $X \sim \text{Gamma}\,(\boldsymbol{\alpha}_1, \boldsymbol{\beta})$, and $X \sim \text{Gamma}\,(\boldsymbol{\alpha}_2, \boldsymbol{\beta})$, then $X{+}Y \sim \text{Gamma}\,(\boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2, \boldsymbol{\beta})$

Beta Distribution

- Beta distribution is a continuous probability distribution defined on the interval $[0, 1]$ making it particularly useful for modeling random variables that represent proportions or probabilities.
- It is characterized by two positive shape parameters, $\boldsymbol{\alpha}$ and β which determine the shape of the distribution.
- The flexibility of the Beta distribution allows it to take on a variety of shapes depending on the values of these parameters, including uniform, U-shaped, and bell-shaped distributions. This versatility makes the Beta distribution a valuable tool in many fields.
- Some areas of application of Beta Distribution include: in Bayesian statistics-Beta distribution is commonly used as a prior distribution for binomial proportions, in project managementit helps model the uncertainty in project completion times. Additionally, in finance-for modeling the distribution of returns on assets and in biology for describing the distribution of allele frequencies in populations among others.

Definition. *The pdf is given by*

$$\mathrm{f}(\mathrm{x}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{\mathrm{x}^{\boldsymbol{\alpha}-1}(1-\mathrm{x})^{\boldsymbol{\beta}-1}}{\mathrm{B}(\boldsymbol{\alpha}, \boldsymbol{\beta})} \qquad \textit{for } \leq \mathrm{x} \leq 1$$

where $B(\boldsymbol{\alpha}, \boldsymbol{\beta})$ is the Beta function and is given as

$$\mathrm{B}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \int_0^1 \mathrm{t}^{\alpha-1}(1-\mathrm{t})^{\beta-1}\mathrm{dt} = \frac{\boldsymbol{\Gamma}(\boldsymbol{\alpha})\boldsymbol{\Gamma}(\boldsymbol{\beta})}{\Gamma(\boldsymbol{\alpha}+\boldsymbol{\beta})}$$

Properties of the Beta distribution

1. The cdf is given by

$$\mathrm{F}(\mathrm{x}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \mathrm{I}_{\mathrm{x}}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{\int_0^{\mathrm{x}} \mathrm{t}^{\alpha-1}(1-\mathrm{t})^{\boldsymbol{\beta}-1}\mathrm{dt}}{\mathrm{B}(\boldsymbol{\alpha}, \boldsymbol{\beta})}$$

2. The mean is

$$\frac{\alpha}{\alpha+\beta}$$

3. The variance is

$$\frac{\alpha \cdot \beta}{(\alpha+\beta)^2 \cdot (\alpha+\beta+1)}$$

Applications of R software in solving problems Involving Probability Distributions

Binomial Distribution

In R, binomial distribution can be used to calculate probabilities and generate random numbers.

Example 15. *Suppose a fair coin is tossed 10 times. What is the probability of getting exactly 6 heads?*

R Code

```
[language $=\mathrm{R}$, caption= R Code for Binomial Distribution]\\
\# Parameters for the binomial distribution\\
n <- 10 \# number of trials\\
$\mathrm{p}<-0.5$ \# probability of success (heads)\\
\# Calculate the probability of getting exactly 6 heads\\
prob <- dbinom(6, size=n, prob=p)\\
prob \# Output: 0.2050781
```

Explanation

- dbinom (x, size, prob): This function computes the probability of observing exactly x successes in n trials with success probability p .

Simulating Binomial Outcomes

Simulate the outcomes of tossing a fair coin 10 times, 1000 times, and plot the distribution of the number of heads.

R Code

```
[language $=\mathrm{R}$, caption=R Code for Simulating Binomial Outcomes]\\
\# Parameters for the binomial distribution\\
n <- 10 \# number of trials (coin tosses)\\
p <- 0.5 \# probability of success (heads)\\
\# Simulate 1000 binomial outcomes\\
set.seed(123) \# for reproducibility\\
sim <- rbinom(1000, size=n, prob=p)\\
\# Plot a histogram of the number of heads\\
hist(sim, breaks=0:n, main="Histogram of Binomial Outcomes", xlab="Number of Heads")
```

Explanation

- rbinom(n, size, prob): This function generates random numbers from the binomial distribution. - hist(): Used to plot the histogram of the simulated results.

Poisson Distribution

Poisson distribution is used to model the number of events occurring in a fixed interval of time or space when these events happen with a known constant rate and independently of each other.

Example Problem The average number of customers arriving at a store per minute is 3 . What is the probability that exactly 5 customers arrive in a given minute?

R Code

```
[language=R, caption=R Code for Poisson Distribution]
# Poisson rate (average number of events)
lambda <- 3 # average number of customers per minute
# Calculate the probability of getting exactly 5 customers
prob <- dpois(5, lambda=lambda)
prob # Output: 0.1008188
```

Explanation

- dpois (x, lambda): This function computes the probability of observing exactly x events when the average rate of occurrence is $\boldsymbol{\lambda}$.

Normal Distribution

In R, functions such as dnorm(), pnorm(), qnorm(), and rnorm() can be used to work with the normal distribution for density, probability, quantile, and random number generation, respectively. Hence, for normal distribution, R can be used to calculate probabilities, generate random numbers and plot the distribution.

Example Problem The heights of a population of men are normally distributed with a mean of 70 inches and a standard deviation of 3 inches. What is the probability that a randomly selected man is between 67 and 73 inches tall?

R Code

```
[language=R, caption=R Code for Normal Distribution]
# Parameters for the normal distribution
mu <- 70 # mean height
sigma <- 3 # standard deviation
# Calculate the probability of height between 67 and 73
prob <- pnorm(73, mean=mu, sd=sigma) - pnorm(67, mean=mu, sd=sigma)
prob # Output: 0.6826895
```

Explanation

- pnorm(q, mean, sd): This function calculates the cumulative probability of a normal distribution up to value q with mean and standard deviation. - The probability that the height is between 67 and 73 is the difference between the cumulative probabilities for 73 and 67 .

Calculating Probability Below a Value

The function pnorm() is used to find the cumulative probability (i.e., $P(X \leq x)$) for a given value from a normal distribution.

R Code

```
[language=R]
# Given: mean = 150, sd = 20, find P(X < 130)
mean <- 150
sd <- 20
x <- 130
# Cumulative probability
p_value <- pnorm(x, mean, sd)
p_value
```

Explanation This code calculates the probability that a random variable from a normal distribution with mean 150 and standard deviation 20 will have a value less than 130. The function pnorm() returns the cumulative probability.

Thus, $P(X < 130) \approx 0.1587$.

Finding a Value Corresponding to a Given Probability

The function qnorm () is used to find the value corresponding to a given cumulative probability.

R Code

```
[language=R]
# Given: mean = 65, sd = 3, find the value below which 90% of the data falls
mean <- 65
sd <- 3
p <- 0.90
# Quantile (inverse of pnorm)
x_value <- qnorm(p, mean, sd)
x_value
```

Explanation This code calculates the value below which 90% of the data falls in a normal distribution with mean 65 and standard deviation 3 using qnorm().

Thus, 90% of the data falls below approximately 68.84.

Probability Between Two Values

To find the probability that a random variable falls between two values, you can calculate the cumulative probabilities for both values and subtract them.

R Code

```
[language=R]
# Given: mean = 100, sd = 15, find P(85 < X < 115)
mean <- 100
sd <- 15
lower_bound <- 85
upper_bound <- 115
# Cumulative probability for both bounds
p_lower <- pnorm(lower_bound, mean, sd)
p_upper <- pnorm(upper_bound, mean, sd)
# Probability between the two values

p_between <- p_upper - p_lower
p_between
```

Explanation This code finds the probability that a random variable from a normal distribution with mean 100 and standard deviation 15 falls between 85 and 115. The difference between the cumulative probabilities at the two points gives the desired probability.

Thus, $P(85 < X < 115) \approx 0.6827$.

Generating Random Numbers from a Normal Distribution

To generate random numbers from a normal distribution, the function rnorm() is used.

R Code

```
[language=R]\\
\# Generate 10 random numbers from a normal distribution with mean = 50, sd = 5 mean <- 50\\
sd <- 5\\
n <- 10\\
\# Random numbers\\
random\_values <- rnorm(n, mean, sd)\\
random\_values
```

Explanation This code generates 10 random values from a normal distribution with mean 50 and standard deviation 5 using rnorm().

Plotting the Normal Distribution

To visualize a normal distribution, you can use the dnorm() function for the density and plot () to display it.

R Code

```
[language $=R$ ]\\
\# Plotting the normal distribution\\
mean <- 0\\
sd <- 1\\
$\mathrm{x}<-\operatorname{seq}(-4,4$, length $=100)$\\
\# Density values\\
$\mathrm{y}<-\operatorname{dnorm}(\mathrm{x}$, mean, sd)\\
\# Plot\\
plot(x, y, type = "l", main = "Standard Normal Distribution", xlab = "x", ylab = "Density",
```

Explanation This code generates a plot of the standard normal distribution (mean = 0, standard deviation = 1) using the dnorm () function to compute the density at each point and plot () to create the graph.

Example: Plotting the Normal Distribution

Plot the probability density function (PDF) of a normal distribution with a mean of 0 and standard deviation of 1.

R Code

```
            R Code for Plotting Normal Distribution
# Define the mean and standard deviation
mu <- 0
sigma <- 1
# Sequence of values for the x-axis
x <- seq (-4, 4, length=100)
# Calculate the PDF values
y <- dnorm(x, mean=mu, sd=sigma)
# Plot the normal distribution
plot(x, y, type="l", main="NormalıDistribution ${ }_{\sqcup}(m e a n=0, \sqcup s d=1) ", x l
```

Explanation

- dnorm(x, mean, sd): This function calculates the PDF for the normal distribution. - The plot() function creates a line graph of the normal distribution.

Exponential Distribution

Exponential distribution is used to model the time between events in a Poisson process, where events occur continuously and independently at a constant rate.

Example Problem

If the average time between arrivals of buses at a station is 10 minutes, what is the probability that the next bus will arrive within 5 minutes?

R Code

```
R Code for Exponential Distribution\\
\# Exponential rate (inverse of the mean time between events) rate <- 1 / 10 \# buses arrive
\# Calculate the probability that the next bus arrives within 5 minutes prob <- pexp(5, rate
prob \#
```

Explanation

- pexp(q, rate): This function computes the cumulative probability up to value q for an exponential distribution with rate 1/ mean.

Simulation of Exponential Random Variables in R

You can simulate Exponential random variables using the following R code. Once you have the variables, you can calculate probabilities, expectation and variance.

R Code

```
# Parameters
lambda <- 1.5 # rate parameter
# Simulation
set.seed(123)
n <- 10000 # number of samples
exp_samples <- rexp(n, rate = lambda)

# Calculating Probabilities
prob_less_than_2 <- pexp(2, rate = lambda)
cat("P(X < 2):", prob_less_than_2, "\n")
# Expectation and Variance
exp_mean <- mean(exp_samples)
exp_variance <- var(exp_samples)
```

```
cat("Exponential Distribution Mean:",
exp_mean, "\n")
cat("Exponential Distribution Variance:",
exp_variance, "\n")
```

Further Examples of Exponential Distribution in R

Example 16. *A machine component has a failure time that follows an exponential distribution with a failure rate of 0.02 failures per hour. What is the probability that the component will last at least 50 hours?*

Solution:

- *The failure rate is given as 0.02 failures per hour. So* $\boldsymbol{\lambda} = 0.02$.
- *For simplicity, use the survival function (the compliment of the CDF) to calculate the probability. That is, the probability that the component lasts at least 50 hours is given by the survival function:*

$$P(X > x) = 1 - F(x; \lambda) = e^{-\lambda \cdot x}$$

- *The probability is therefore*

$$\begin{aligned} \mathrm{P}(X > 50) &= \mathrm{e}^{-0.02 \times 50} \\ &= \mathrm{e}^{-1} = 0.3679. \end{aligned}$$

- So the probability that the component will last at least 50 hours is approximately 36.79%.
- In R this is implemented as

R Code

```
# Given failure rate (lambda)
lambda <- 0.02
# Time period
t <- 50
# Calculate the probability that the component will last at
# least 50 hours
prob_at_least_50 <- exp(-lambda * t)
cat("The probability that the component will last at
least 50 hours is:", prob_at_least_50, "\n")
```

Example 17. *A light bulb has an average lifetime that follows an exponential distribution with a mean lifetime of 2000 hours. Calculate the expectation and variance of the lifetime of the light bulb.*

Solution:

- *The mean lifetime* $\boldsymbol{\mu}$ *is given as 2000 hours.*
- *The rate parameter* $\boldsymbol{\lambda}$ *is the reciprocal of the mean*

$$\lambda = \frac{1}{\mu} = \frac{1}{2000} \text{ per hour}$$

- *The variance is given as*

$$\sigma^2 = \left(\frac{1}{\frac{1}{2000}}\right)^2 = 4,000,000 \text{ hours}$$

- *In R this is implemented as*

R Code

```
# Given mean lifetime
mean_lifetime <- 2000
# Calculate the rate parameter (lambda)
lambda <- 1 / mean_lifetime
# Calculate Expectation (Mean)
exp_mean <- mean_lifetime
cat("Exponential Distribution Mean:", exp_mean, "hours\n")
# Calculate Variance
exp_variance <- 1 / (lambda^2)
cat("Exponential Distribution Variance:", exp_variance, "(hours)^2\n")
```

Example 18. *Suppose that the time between arrivals of buses at a certain bus stop follows an exponential distribution with a rate parameter* $\boldsymbol{\lambda} = 3$ *buses per hour. Calculate the expectation and variance of the time between arrivals.*

Solution:

- *You have that* $\boldsymbol{\lambda} = 3$
- *The mean is therefore*

$$\mu = \frac{1}{\lambda} = \frac{1}{3} \text{ hours}$$

- *The variance is*

$$\sigma^2 = \frac{1}{3^3} = \frac{1}{9} \text{ hours }^2$$

- ln R *this is solved as:*

```
\# Parameter\\
lambda <- 3\\
\# Calculate Expectation (Mean)\\
exp\_mean <- 1 / lambda\\
cat("Exponential Distribution Mean:", exp\_mean, "hours\textbackslash n")\\
\# Calculate Variance\\
exp\_variance <- 1 / (lambda\^{}2)\\
cat("Exponential Distribution Variance:", exp\_variance, "(hours) 2 2 n")
```

Simulation of Gamma Random Variables in R

You can simulate Exponential random variables using the following R code. Once you have the variables, you can calculate probabilities, expectation and variance.

```
# Parameters
alpha <- 3 # shape parameter
beta <- 2 # rate parameter
# Simulation
set.seed(123)
gamma_samples <- rgamma(n, shape = alpha, rate = beta)
# Calculating Probabilities
prob_between_1_and_2 <- pgamma(2, shape = alpha,
rate = beta) - pgamma(1, shape = alpha, rate = beta)
cat("P(1 < X < 2):", prob_between_1_and_2, "\n")
# Expectation and Variance
gamma_mean <- mean(gamma_samples)
gamma_variance <- var(gamma_samples)
cat("Gamma Distribution Mean:", gamma_mean, "\n")
cat("Gamma Distribution Variance:", gamma_variance, "\n")
```

Example 19. *Suppose the time between customer arrivals at a service center follows a Gamma distribution with shape parameter $\boldsymbol{\alpha} = 2$ and rate parameter $\boldsymbol{\beta} = 0.5$ hours. What is the probability that the time between two consecutive customer arrivals is between 1 and 2 hours?*

Solution

- *Given $\alpha = 2$ and $\beta = 0.5$*
- *This is implemented in R as*

```
# Parameters
alpha <- 2
beta <- 0.5
```

```
# Calculate CDF values
cdf_2 <- pgamma(2, shape = alpha, rate = beta)
cdf_1 <- pgamma(1, shape = alpha, rate = beta)
# Calculate the probability P(1 < X < 2)
prob_1_to_2 <- cdf_2 - cdf_1
# Print the result
cat("The probability that the time between two
consecutive customer arrivals is between 1 and 2
hours is:", prob_1_to_2, "\n")
```

Example 20. *Suppose the time until failure for a particular type of machine part follows a Gamma distribution with shape parameter $\boldsymbol{\alpha} = 3$ and rate parameter $\boldsymbol{\beta} = 2$ (failures per hour). Calculate the expectation and the variance of the time until failure.*

Solution

- *Since $\boldsymbol{\alpha} = 3, \boldsymbol{\beta} = 2$, the mean is*

$$\mu = \frac{\alpha}{\beta} = \frac{3}{2} = 1.5 \ hours$$

- *The variance is given by:*

$$\mathrm{Var}(X) = \frac{\alpha}{\beta^2} = \frac{3}{4} = 0.75 \ hours$$

- *In R, this is implemented as*

```
\# Parameters\\
alpha <- 3\\
beta <- 2\\
\# Calculate Expectation (Mean)\\
gamma\_mean <- alpha / beta\\
cat("Gamma Distribution Mean:", gamma\_mean, "hours\textbackslash n")\\
\# Calculate Variance\\
gamma\_variance <- alpha / (beta\^{}2)\\
```

Simulation of Beta Random Variables in R

You can simulate Exponential random variables using the following R code. Once you have the variables, you can calculate probabilities, expectation and variance.

```
# Parameters
alpha <- 2 # shape parameter
beta <- 5 # shape parameter
# Simulation
set.seed(123)
beta_samples <- rbeta(n, shape1 = alpha, shape2 = beta)
# Calculating Probabilities
prob_less_than_0.3 <- pbeta(0.3, shape1 = alpha, shape2 = beta)
cat("P(X < 0.3):", prob_less_than_0.3, "\n")
# Expectation and Variance
beta_mean <- mean(beta_samples)
beta_variance <- var(beta_samples)
cat("Beta Distribution Mean:", beta_mean, "\n")
cat("Beta Distribution Variance:", beta_variance, "\n")
```

Example 21. *Suppose we have a Beta distribution with shape parameters $\alpha = 2$ and $\beta = 5$. Calculate the following*

1. *The mean and the variance*

   Solution:

   - *Using the formulae for mean and variance, we have*

$$\begin{aligned}\mu &= \frac{\alpha}{\alpha+\beta} \\ &= \frac{2}{7} = 0.2857 \\ \sigma^2 &= \frac{\alpha \cdot \beta}{(\alpha+\beta)^2 \cdot (\alpha+\beta+1)} \\ &= \frac{10}{49 \cdot 8} = 0.0255.\end{aligned}$$

   - *This can be implemented in R as:*

```
# Parameters
alpha1 <- 2
beta1 <- 5
# Calculate Mean
beta_mean1 <- alpha1 / (alpha1 + beta1)
cat("Beta Distribution Mean:", beta_mean1, "\n")
# Calculate Variance
beta_variance1 <- (alpha1 * beta1) /
((alpha1 + beta1)^2 * (alpha1 + beta1 + 1))
cat("Beta Distribution Variance:", beta_variance1, "\n")
```

Congratulations!!! You are almost finishing this Module. However, before you proceed, take some minutes to do a recap of what you have learned so far.

Conclusion. *Applications of Probability Distributions in Artificial Intelligence You are at the last section of this Module. In this section, you are going to learn about how the concept of Probability Distributions is applied in Artificial Intelligence.*

Probability distributions are key in Al especially when it comes to modeling uncertainty and making inferences and predictions based on some data among other uses. Some key areas are explained herein:

Bayesian Networks

Bayesian network involves using a directed acyclic graph (DAG) to graphically model a set of variables and their conditional dependencies. In this model, each node in the graph represents a random variable, while the edges represent conditional dependencies between these variables and hence, the need for knowledge of conditional probability distributions.

Application

In decision-making, especially under uncertainty, Bayesian networks can, for instance, be used to do the following:

- **In medical diagnosis:** Modeling the relationship between diseases and symptoms, helping doctors infer the likelihood of diseases given observed symptoms.
- **In autonomous systems:** Helping robots to make decisions based on uncertain sensor data. In this case, probability distribution of a variable X given its parent variables $\mathrm{Pa}(X)$ is usually modeled as:

$$\mathrm{P(X \mid Pa(X))}$$

The joint probability distribution over all variables is the product of the local conditional distributions:

$$\mathrm{P\left(X_1, X_2, \ldots, X_n\right) = \prod_{i=1}^{n} P\left(X_i \mid Pa\left(X_i\right)\right)}$$

That is, the joint distribution is the product of the marginal distributions

Bayesian Inferencing in AI Models

Bayesian inference involves use of probability concept to define and/or estimate some beliefs or hypothesis. In other words, Bayesian inference uses statistical technique to update the probability estimate of a hypothesis as more evidence

prevails.
In Al, this concept is used to update models and predictions based on new data, particularly in cases of sparse or noisy data.

Application

In Al, the concept of probability distributions can be applied through Bayesian inferencing in:

- Bayesian Neural Networks (BNNs): These models incorporate uncertainty in their weights by assigning probability distributions to them.
- Model Updating: Al systems such as recommendation engines use Bayesian inference to update user preferences as more interactions are observed.
  In Bayesian inference, the posterior distribution is updated based on prior distribution and likelihood:

$$P(\boldsymbol{\theta} \mid \mathrm{X})=\frac{\mathrm{P}(\mathrm{X} \mid \boldsymbol{\theta}) \mathrm{P}(\boldsymbol{\theta})}{\mathrm{P}(\mathrm{X})}$$

where $P(\theta \mid X)$ is the posterior probability of the model parameters $\boldsymbol{\theta}$ given data $X, P(\theta)$ is the prior, and $\mathrm{P}(\mathrm{X} \mid \theta)$ is the likelihood.

Hidden Markov Models (HMMs)

Hidden Markov Models are probabilistic models used to represent systems that evolve over time. They consist of hidden states, where each state follows a probability distribution, and the model attempts to infer the sequence of hidden states based on observed data. HMMs are characterized by two distributions: the transition probability distribution (between hidden states) and the observation probability distribution (from hidden states to observations).

Application

HMMs are widely used in:

- Natural Language Processing (NLP): For tasks such as part-of-speech tagging, speech recognition, and language modeling.
- Time Series Analysis: For modeling stock prices, weather patterns, or any other timedependent data.

For an HMM, the probability of transitioning from state s_i to s_j is given by:

$$P\left(S_{t+1}=s_j \mid S_t=s_i\right)=A_{ij}$$

The probability of observing a particular output o_t given a state s_t is given by the emission probability:

$$P\left(O_t=o_t \mid S_t=s_t\right)=B_{s_t}\left(o_t\right)$$

Gaussian Mixture Models (GMMs)

A Gaussian Mixture Model (GMM) is a probabilistic model that assumes that all data points are generated from a mixture of several Gaussian distributions, each with its own mean and variance. GMMs are key in clustering and density estimation tasks.

Application

In Al, GMMs are used for:

- Clustering: For unsupervised learning tasks such as image segmentation, GMMs help in modeling clusters of similar data points.
- Anomaly Detection: GMMs can be used to model normal behavior, and deviations from the model can be flagged as anomalies.

The probability distribution of a GMM with k Gaussian components is given by:

$$\mathrm{P}(\mathrm{x})=\sum_{\mathrm{i}=1}^{\mathrm{k}} \mathrm{w}_{\mathrm{i}} \mathcal{N}\left(\mathrm{x} \mid \boldsymbol{\mu}_{\mathrm{i}}, \boldsymbol{\Sigma}_{\mathrm{i}}\right)$$

where w_{i} is the weight of the i -th Gaussian component, $\mathcal{N}\left(\mathrm{x} \mid \boldsymbol{\mu}_{\mathrm{i}}, \boldsymbol{\Sigma}_{\mathrm{i}}\right)$ is the probability density function of a Gaussian distribution with mean $\boldsymbol{\mu}_{\mathrm{i}}$ and covariance $\boldsymbol{\Sigma}_{\mathrm{i}}$.

Reinforcement Learning (RL)

In reinforcement learning (RL), an agent learns to take actions in an environment to maximize cumulative rewards.
Probabilistic models such as Markov decision processes (MDPs) are used to represent the environment and the uncertainty associated with the outcomes of actions.

Application

Probability distributions are used in RL to:

- Model Uncertainty in Actions: In stochastic environments, the effect of an action is not deterministic. Instead, it follows a probability distribution.
- Value Function Estimation: Probability distributions help estimate the expected future reward for different actions.

For an MDP, the probability of transitioning from state s to state s' after taking action a is given by the transition probability:

$$P\left(s' \mid s, a\right)$$

The goal is to find a policy π(a | s) that maximizes the expected cumulative reward.

Variational Autoencoders (VAEs)

Variational autoencoders are generative models that use probability distributions to learn low-dimensional latent representations of data.

VAEs assume that data is generated from a distribution in the latent space, and they optimize the parameters of the distribution using a technique called variational inference.

Application

VAEs are widely used in:

- Generative Modeling: VAEs generate new data samples from the learned distribution, making them useful in image and text generation tasks.
- Representation Learning: VAEs learn compact representations of high-dimensional data, which can be used for tasks such as data compression or anomaly detection.
  The VAE optimizes the following evidence lower bound (ELBO) on the log-likelihood:

$$\log P(x) \geq \mathbb{E}_{q(z|x)}[\log P(x \mid z)] - D_{KL}(q(z \mid x) \| p(z))$$

where q(z | x) is the approximate posterior distribution, P(x | z) is the likelihood, and $\mathrm{D_{KL}}$ is the Kullback-Leibler divergence between the approximate and true posterior distributions.

Having learned about some common ways in which the concept of Probability Distributions is applied in AI, I believe you now appreciate why this module is important to you. The video clip below will further highlight some applications of Probability Distributions in Al.

Video Visit the URL below to view a video:

https://www.youtube.com/embed/Gts-fIsRUB8